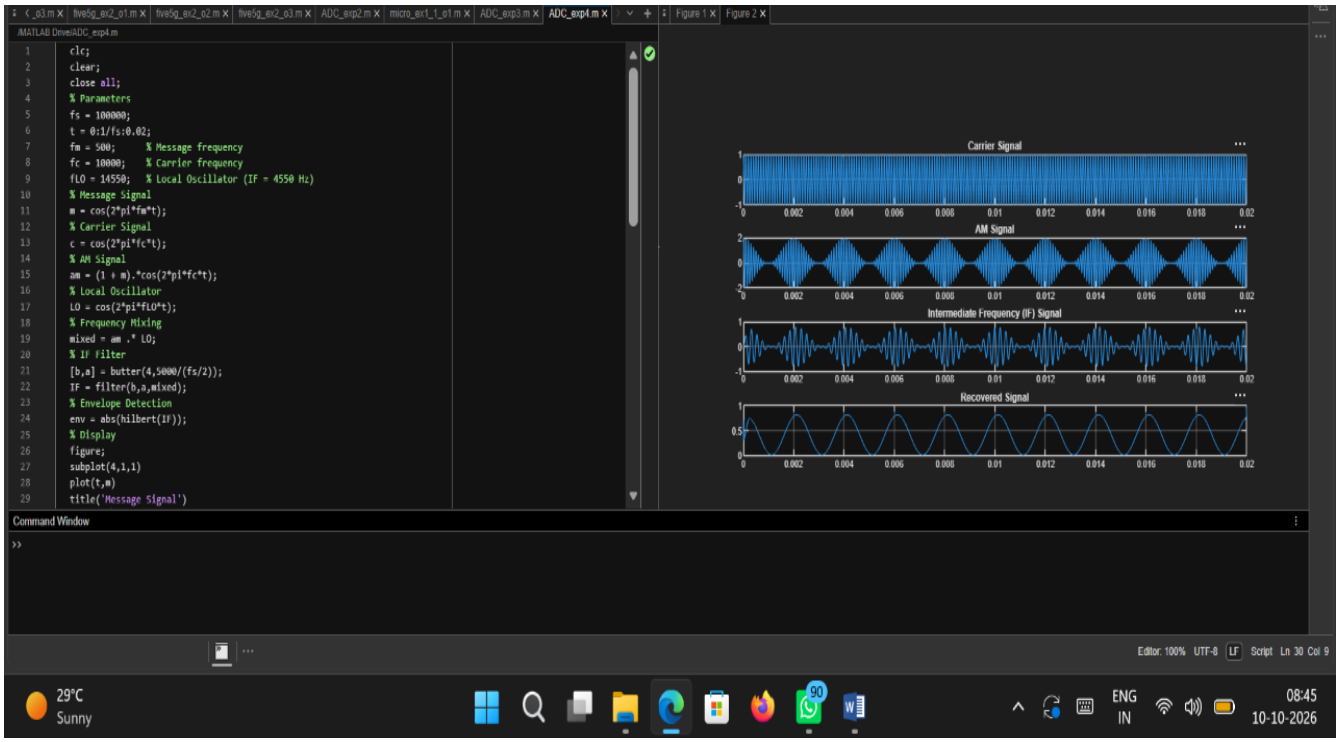
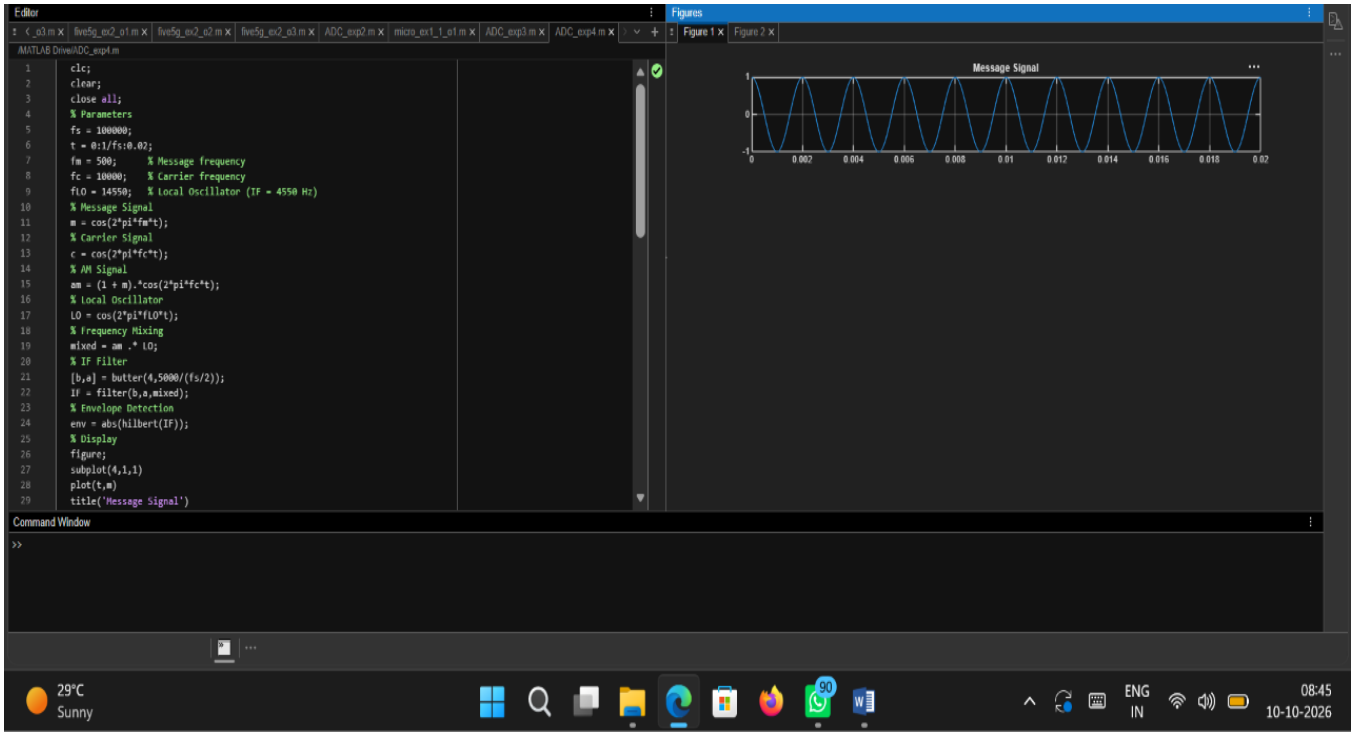
Ex.No:4: MATLAB SIMULATION Of SUPERHETERODYNE RECEIVER FOR AM COMMUNICATION SYSTEM WITH FREQUENCY MIXING AND FILTERING

Objective 1:



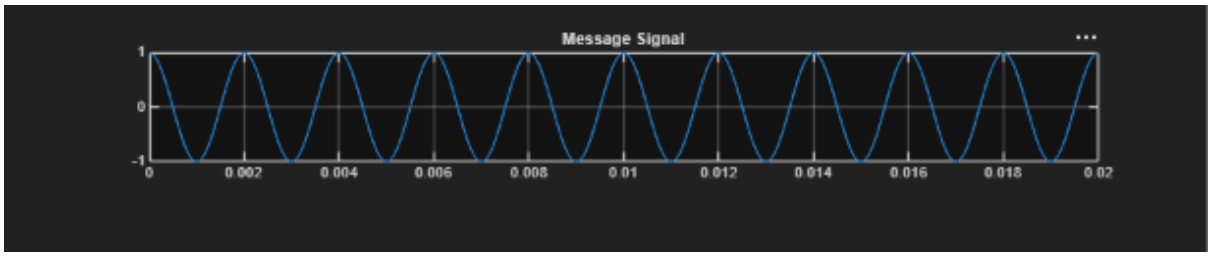


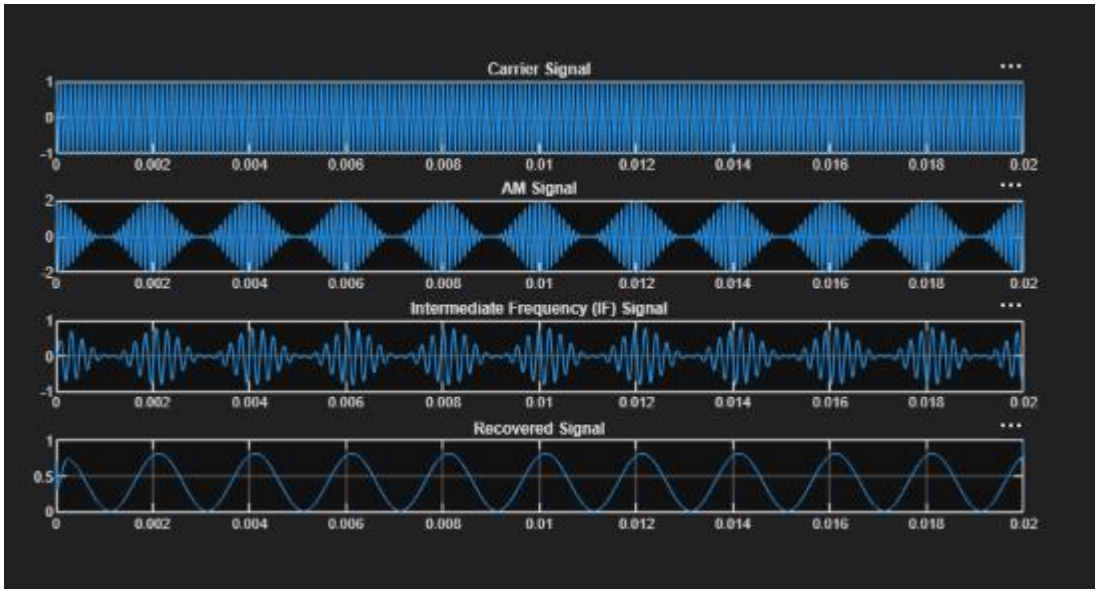
Code:

```
clc;
clear;
close all;
% Parameters
fs = 100000;
t = 0:1/fs:0.02;
fm = 500;      % Message frequency
fc = 10000;    % Carrier frequency
fLO = 14550;   % Local Oscillator (IF = 4550 Hz)
% Message Signal
m = cos(2*pi*fm*t);
% Carrier Signal
c = cos(2*pi*fc*t);
% AM Signal
am = (1 + m).*cos(2*pi*fc*t);
% Local Oscillator
LO = cos(2*pi*fLO*t);
% Frequency Mixing
mixed = am .* LO;
% IF Filter
[b,a] = butter(4,5000/(fs/2));
IF = filter(b,a,mixed);
% Envelope Detection
```

```
env = abs(hilbert(IF));
% Display
figure;
subplot(4,1,1)
plot(t,m)
title('Message Signal')
grid on
figure;
subplot(5,1,2)
plot(t,c)
title('Carrier Signal')
grid on
subplot(5,1,3)
plot(t,am)
title('AM Signal')
grid on
subplot(5,1,4)
plot(t,IF)
title('Intermediate Frequency (IF) Signal')
grid on
subplot(5,1,5)
plot(t,env)
title('Recovered Signal')
grid on
```

Output:


Message Signal
1
0
-1
0
0.002
0.004
0.006
0.008
0.01
0.012
0.014
0.016
0.018
0.02


Carrier Signal
AM Signal
Intermediate Frequency (IF) Signal
Recovered Signal
0.5
0.002
0.004
0.006
0.008
0.01
0.012
0.014
0.016
0.018
0.02